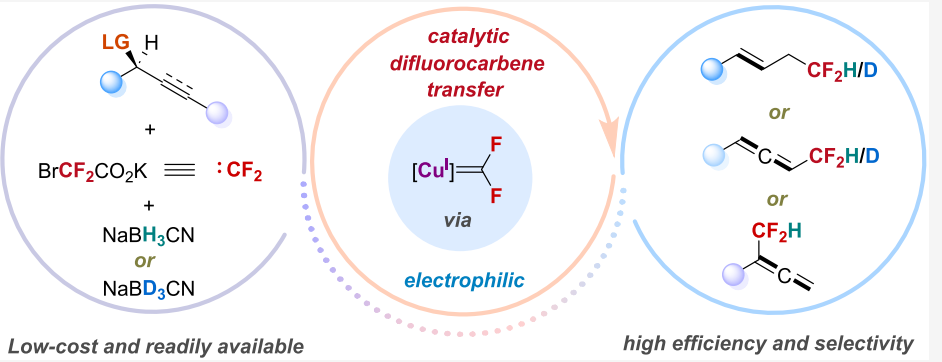
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Copper Difluorocarbene Enables Catalytic Difluoromethylation

Xin Zeng, Jimin Yang, Wei Deng, Xiao-Tian Feng, Hai-Yang Zhao, Lanfeng Wei, Xiao-Song Xue,* and Xingang Zhang*

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ABSTRACT: Despite the synthetic versatility of difluorocarbene, its high reactivity severely regulates widespread applications of difluorocarbene in organic synthesis. Here, we report a copper difluorocarbene-involved catalytic coupling, representing a new mode of the difluoromethylation reaction. This method allows difluoromethylation of a wide range of readily available allyl/propargyl electrophiles with $NaBH_3CN$ and low-cost difluorocarbene precursor $BrCF_2CO_2K$, featuring high cost-efficiency, high stereo- and regioselectivities, and high functional group tolerance, even with complex drug-like molecules. Applying the method led to the efficient synthesis of deuterated difluoromethylated compounds of medicinal interest. The resulting difluoromethylated allyl and allenyl products can serve as versatile synthons for diverse transformations, rendering the approach attractive for synthesizing complex fluorinated structures. Experimental mechanistic studies and computational calculations reveal that the formation of a difluoromethylcopper(I) intermediate through the nucleophilic attack of boron hydride on the copper(I) difluorocarbene is the key step in the reaction.

INTRODUCTION

Difluorocarbene has received increasing attention due to its synthetic versatility in forming two bonds to its carbon centers.[1] This distinct property significantly expands the chemical space to construct new fluorinated structures for modern drug discovery[2] and advanced functional materials.[3] However, the high reactivity of difluorocarbene severely regulates its widespread synthetic applications. As a result, the reaction types of difluorocarbene are very limited.[3a,4] Using metal to modulate the reactivity of difluorocarbene is a promising strategy to overcome this limitation. In this context, efforts have been made to synthesize and characterize metal difluorocarbene complexes over the past 40 years.[5] However, the use of metal difluorocarbene complexes for catalytic synthesis remains a long-standing challenge due to the lack of catalytic reactivity of the isolated metal difluorocarbene complexes.[5,6]

Recently, we developed a palladium difluorocarbene-involved catalytic coupling,[7] enabling the efficient coupling of difluorocarbene with aryl borons/alkynes and proton.[8] In this process, a palladium(0) difluorocarbene $[Pd^0]{=}CF_2$ complex was isolated and showed strong nucleophilicity.[8c] This complex can be protonated by H_2O to form a difluoromethylpalladium complex $[XPd^{II}(L_n)CF_2H]$, making it a key intermediate in the palladium-catalyzed difluoromethylation reactions (Scheme 1A). We also recently isolated and characterized copper(I) difluorocarbene $[Cu^I]{=}CF_2$ complex.[9] In contrast to the $[Pd^0]{=}CF_2$ complex, the $[Cu^I]{=}CF_2$ complex exhibits electrophilicity (Scheme 1A), even though these two metals possess the same d electron count. This electrophilic $[Cu^I]{=}CF_2$ species can be attacked by silyl enol ethers, enabling the catalytic modular synthesis of fluorinated compounds.[9] This intriguing finding inspired us to question whether $[Cu^I]{=}CF_2$ can react with a hydride to generate difluoromethylcopper complex $[Cu^I(L_n)CF_2H]$ in a polar

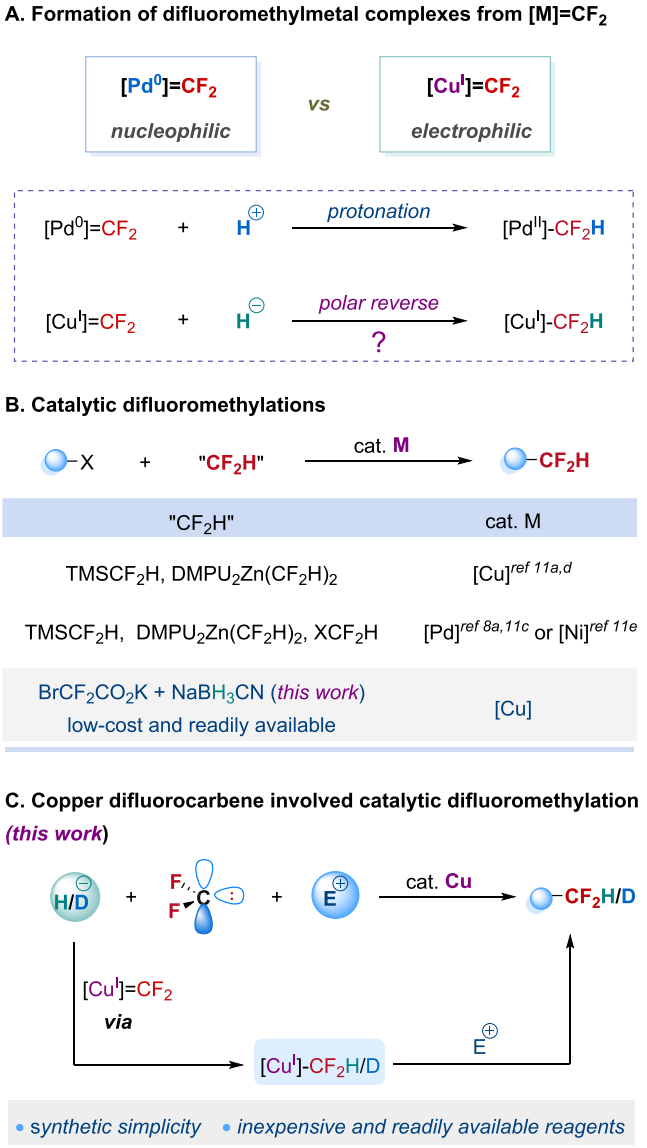
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Scheme 1. Catalytic Difluoromethylations



reverse manner compared to palladium difluorocarbene chemistry (Scheme 1A), thereby paving a new way to the formation of a difluoromethyl metal complex for catalytic difluoromethylation reactions. However, copper difluorocarbene chemistry remains in its infancy despite the prospect of a new retrosynthetic paradigm to create valuable fluorinated compounds. Although impressive progress has been made in the catalytic difluoromethylation reactions,[10] most developed methods rely on expensive difluoromethylating reagents or metal catalysts (Scheme 1B).[11] Developing cost-efficient catalytic difluoromethylation reactions from inexpensive and widely available feedstocks remains in high demand because of the unique properties of the difluoromethyl group that plays a vital role in drug discovery and development.[12] Copper is an earth-abundant and low-cost metal catalyst. However, the cost-efficient copper-catalyzed difluoromethylation reactions still need to be improved due to the lack of inexpensive and readily available difluoromethylating reagents (Scheme 1B).[11a,b,d] To the best of our knowledge, the use of a hydride to construct the difluoromethyl group for the catalytic reaction has yet to be reported.

Here, we report a new mode of difluoromethylation reaction, copper difluorocarbene-involved catalytic difluoromethylation (Scheme 1C). This innovative reaction allows difluoromethylation of a wide range of readily available allyl/propargyl electrophiles with low-cost difluorocarbene precursor $BrCF_2CO_2K$[13,14] and $NaBH_3CN$, providing cost-efficient access to synthetically versatile difluoromethyl allyl and allenyl compounds. Applying the method led to the efficient synthesis of deuterated difluoromethylated compounds of medicinal interest. Experimental mechanistic studies and computational calculations revealed that the formation of the key intermediate $[Cu^I]-CF_2H$ through the nucleophilic attack of boron hydride to $[Cu^I]=CF_2$ is the key step in this catalytic reaction.

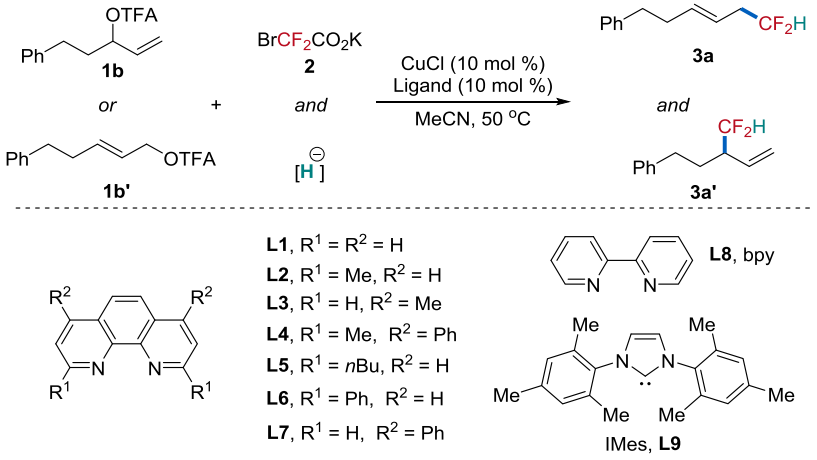
RESULTS AND DISCUSSION

Optimization of the Reaction Conditions. To test our hypothesis, initially, we chose readily available branched allyl acetate **1a** as the model substrate and low-cost $BrCF_2CO_2K$ **2** as the difluorocarbene precursor. After extensive efforts using various hydride sources, including silanes and boron hydrides, we failed to achieve the desired product **3a** (Table S1). Replacing the acetate group with a more reactive leaving group, trifluoroacetate, benefits the reaction efficiency (Table 1). A 25% yield of linear difluoromethylated product **3a** was obtained when the reaction mixture of **1b** (1.0 equiv) and **2** (2.0 equiv) was treated with triethylsilane (Et_3SiH, 1.5 equiv) and K_3PO_4 in the presence of CuCl (10 mol %) and 1,10-phen **L1** (10 mol %) in MeCN at 50 °C (entry 1). A survey of the hydride sources (entries 2−6) showed $NaBH_3CN$ was the optimal choice, providing **3a** in 73% yield with high stereoselectivity (E/Z = 15:1) without observation of branched product **3a′** (entry 6). The reaction was not sensitive to the ligand, as phen-, bpy-, and NHC-based ligands produced **3a** (entries 7−13). Proper steric ligand 2,9-diMe-1,10-phen **L2** could improve the yield to 82% with an excellent E/Z ratio (E/Z >99:1, entry 7). The comparable yields were also obtained using 2,9-diPh-1,10-phen **L6** (76%, entry 11), 2,2′-bpy **L8** (84%, entry 13), and IMes **L9** (82%, entry 14), with slightly lower E/Z ratios observed. However, bidentate phosphine ligands, such as Xantphos and DPEphos, diminished the yields (Table S4). Finally, decreasing the reaction concentration and using 5 mol % CuCl could provide **3a** in 85% isolated yield (E/Z >99:1, entry 15). Linear trifluoroacetate **1b′** was also applicable to the reaction, with a lower yield (35%) obtained (entry 16), probably due to the easier oxidative addition of copper(I) with branched allyl trifluoroacetates **1b** via an S_N2' oxidative addition pathway.[15] No product was observed without copper salt (entry 17). Yet the absence of ligand provided **3a** in good yield (68%) with lower E/Z selectivity (E/Z = 16:1, entry 18), demonstrating the essential role of copper salt in promoting the reaction.

Substrate Scope Analysis and Synthetic Applications. With the viable reaction conditions in hand, we examined the substrate scope of this copper difluorocarbene-involved catalytic difluoromethylation reaction (Scheme 2). Various branched aliphatic allyl trifluoroacetates **1** underwent coupling efficiently with excellent linear selectivity (**3a−3i, 3n, 3r**).[16] The length of the carbon side chain did not interfere with the reaction efficiency; even the substrates bearing additional alkene moiety (**3b−3d**) were competent coupling partners without observation of [2 + 1] cycloaddition side products generated between difluorocarbene and carbon−carbon double bond.[4c] Aryl-substituted allyl trifluoroacetates were also

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Table 1. Representative Results for the Optimization of the Reaction Conditions[a]



entry	[H⁻]	ligand	3a (%)[b]	E/Z[b]
1[c]	Et_3SiH	L1	25	
2[c]	$(EtO)_3SiH$	L1	nd	
3[c]	Ph_3SiH	L1	nd	
4	HBpin	L1	16	
5	$NaBH_4$	L1	19	
6	$NaBH_3CN$	L1	73	15:1
7	$NaBH_3CN$	L2	82	>99:1
8	$NaBH_3CN$	L3	58	9:4
9	$NaBH_3CN$	L4	71	>99:1
10	$NaBH_3CN$	L5	81	>99:1
11	$NaBH_3CN$	L6	76	18:1
12	$NaBH_3CN$	L7	70	13:3
13	$NaBH_3CN$	L8	84	41:1
14[d]	$NaBH_3CN$	L9	82	20:1
15[e]	$NaBH_3CN$	L2	87 (85)	>99:1
16[e,f]	$NaBH_3CN$	L2	35	>99:1
17[g]	$NaBH_3CN$	L2	nd	
18	$NaBH_3CN$	none	68	16:1

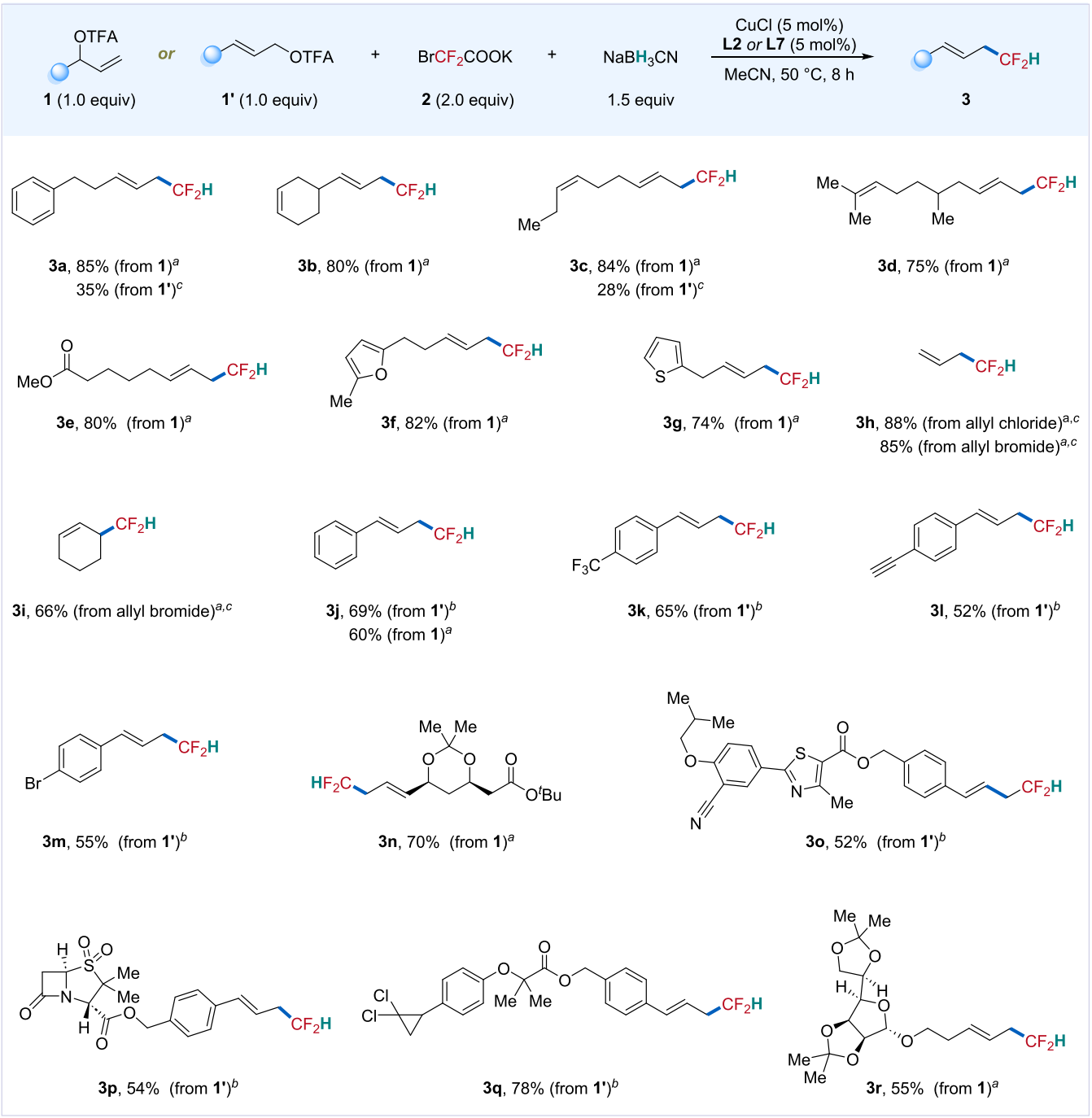
[a]Reaction conditions (unless otherwise specified): **1b** (0.2 mmol, 1,0 equiv), **2** (0.4 mmol, 2.0 equiv), [H⁻] (1.5 equiv), MeCN (2 mL), 8 h. [b]Determined by ^{19}F NMR using fluorobenzene as an internal standard. The number in the parentheses is the isolated yield. [c]1.0 equiv of K_3PO_4 was used. [d]Using IMesCuCl instead of CuCl/**L2**. [e]Using 5 mol % CuCl and **L2**, MeCN (4 mL). [f]Using linear substrate **1b′** instead of **1b**. [g]Without copper salt.

applicable to the reaction (**3j**−**3m**, **3o**−**3q**), in which **L7** and **NaI** were needed (Table S7). The role of NaI is to slow down the decomposition of the difluorocarbene precursor $BrCF_2CO_2K$, thus facilitating the catalytic cycle.[9] In contrast to linear aliphatic allyl trifluoroacetates **1′** that afforded their corresponding products in low yields (**3a**, **3c**), branched and linear aromatic allyl trifluoroacetates were more reactive, providing the difluoromethylated products in good yields (**3j**). The reaction exhibited a high functional group tolerance. Important synthetic handles, such as carboxylic acid ester (**3e**), furan (**3f**), thiophene (**3g**), alkyne (**3l**), and aryl bromide (**3m**), were compatible with the reaction conditions, offering good opportunities for derivatization. Notably, substrates derived from complex molecules, such as protected chiral diol (**3n**), febuxostat (**3o**), sulbactam (**3p**), ciprofibrate (**3q**), and carbohydrate (**3r**), were amenable to this copper-catalyzed process, demonstrating the potential application of the current approach in modern drug discovery. The reaction can also be extended to allyl chlorides and bromides, as exemplified by the efficient synthesis of **3h**. Furthermore, the cyclic allyl electrophile was also a suitable substrate, providing difluoromethylated product **3i** efficiently. However, aryl and alkenyl trifluoroacetates were not applicable to the reaction, which would be addressed in the future.

The reaction was not restricted to allyl substrates, as propargyl electrophiles were also suitable coupling partners with **L6** as the ligand, providing a wide range of difluoromethylated allenes with high efficiency and selectivity (Scheme 3). Although extensive efforts have been made in the preparation of allenes,[17] scarce examples of efficient synthesis of difluoromethylated allenes have been reported.[16b,18] Given the synthetic versatility of allenes[17c−g] and the unique properties of the difluoromethyl group in medicinal chemistry, developing efficient methods to access these valuable fluorinated structures is of great interest. We found that the regioselectivity of the difluoromethylation could be controlled by the propargyl substrates: branched propargyl sulfonates provided "linear" products (**6a**−**6k**), while linear propargyl sulfonates or chlorides afforded "branched" allenes as the major products (**7a**−**7d**), thereby providing a synthetically versatile route to difluoromethylated allenes. A range of "linear" difluoromethylated allenes could be efficiently

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Scheme 2. Copper Difluorocarbene-Involved Catalytic Difluoromethylation of Allyl Electrophiles



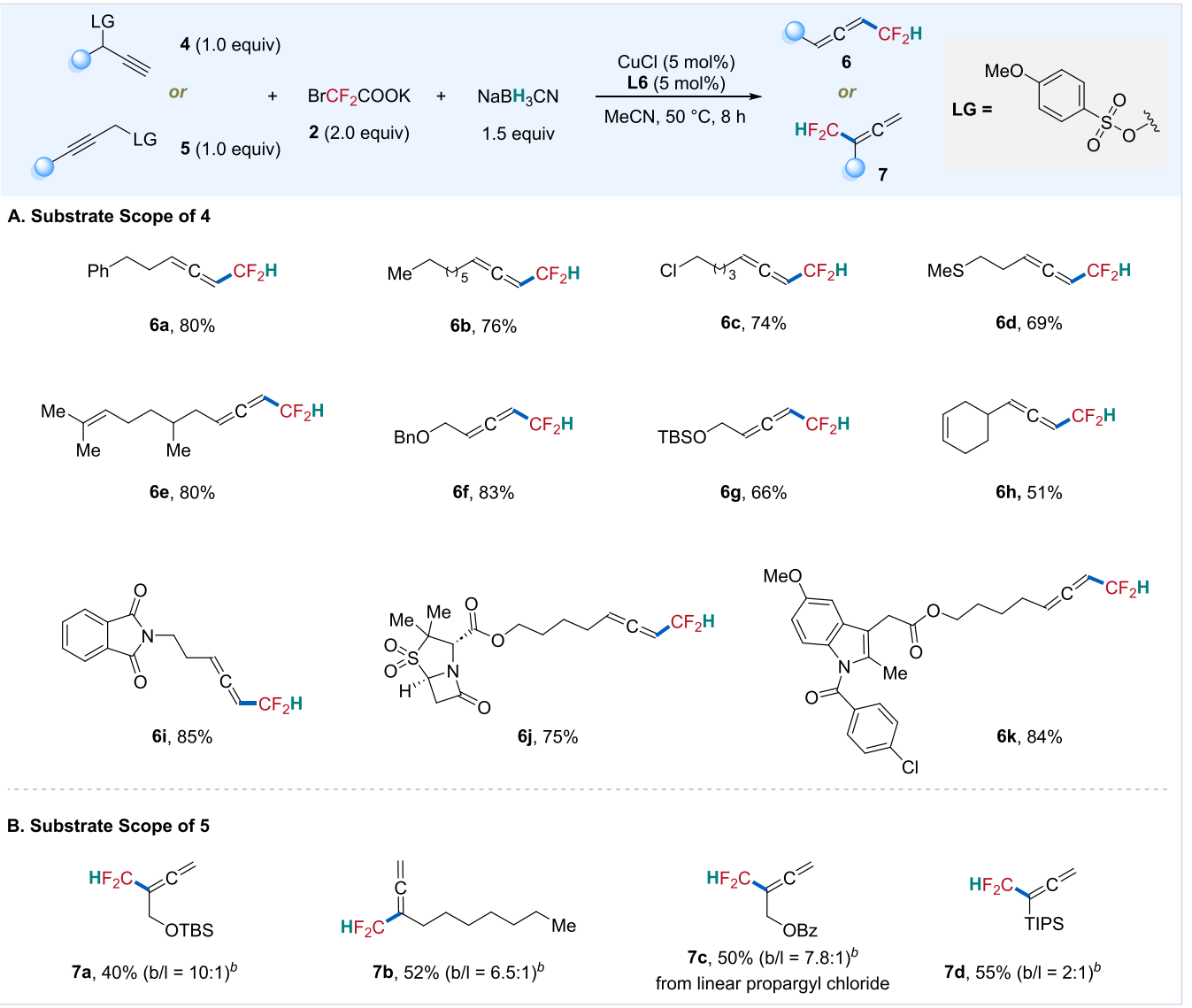
[a]Condition A: **1** or **1′** (0.4 mmol, 1.0 equiv), **2** (2.0 equiv), $NaBH_3CN$ (1.5 equiv), CuCl (5 mol %), **L2** (5 mol %), MeCN (8 mL). [b]Condition B: **1′** (0.4 mmol, 1.0 equiv), **2** (2.0 equiv), $NaBH_3CN$ (1.5 equiv), CuCl (5 mol %), **L7** (5 mol %), NaI (0.1 equiv), and MeCN (8 mL). [c]Yield was determined by ^{19}F NMR.

achieved through this approach (**6a**−**6k**); even drug-containing substrates, including sulbactam (**6j**) and indomethacin (**6k**), underwent the copper-catalyzed process smoothly, with good yields obtained, thus demonstrating the advancement of this protocol. High "branched-regioselectivity" was obtained when linear proparyl sulfonates were used (**7a**−**7c**). Notably, even sterically hindered triisopropylsilyl-substituted substrate still afforded the "branched" product efficiently, with a 2:1 ratio obtained (**7d**). Additionally, linear propargyl chloride was also applicable to the reaction, providing **7c** in a good yield and high regioselectivity. Benzylic electrophiles were also suitable substrates with CuBr as the catalyst, 3,4,7,8-tetraMe-1,10-phen (**L16**) as the ligand, and HBpin as the hydride source (Table S9). For instance, benzyl bromide could yield the corresponding difluoromethylated product **8** in 62% yield (Table S9).

Remarkably, this three-component reaction could be easily applied to the preparation of deuterated difluoromethylated compounds with $NaBD_3CN$ under the standard reaction conditions (Scheme 4). Generally, allyl trifluoroacetates and

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Scheme 3. Copper Difluorocarbene-Involved Catalytic Difluoromethylation of Propargyl Electrophiles[a]



[a]**4** or **5** (0.4 mmol, 1.0 equiv), **2** (2.0 equiv), $NaBH_3CN$ (1.5 equiv), CuCl (5 mol %), **L6** (5 mol %), MeCN (4 mL). [b]The yield is based on the mixture of linear and branched products. l, linear; b, branched. The ratio of l:b was determined by ^{19}F NMR.

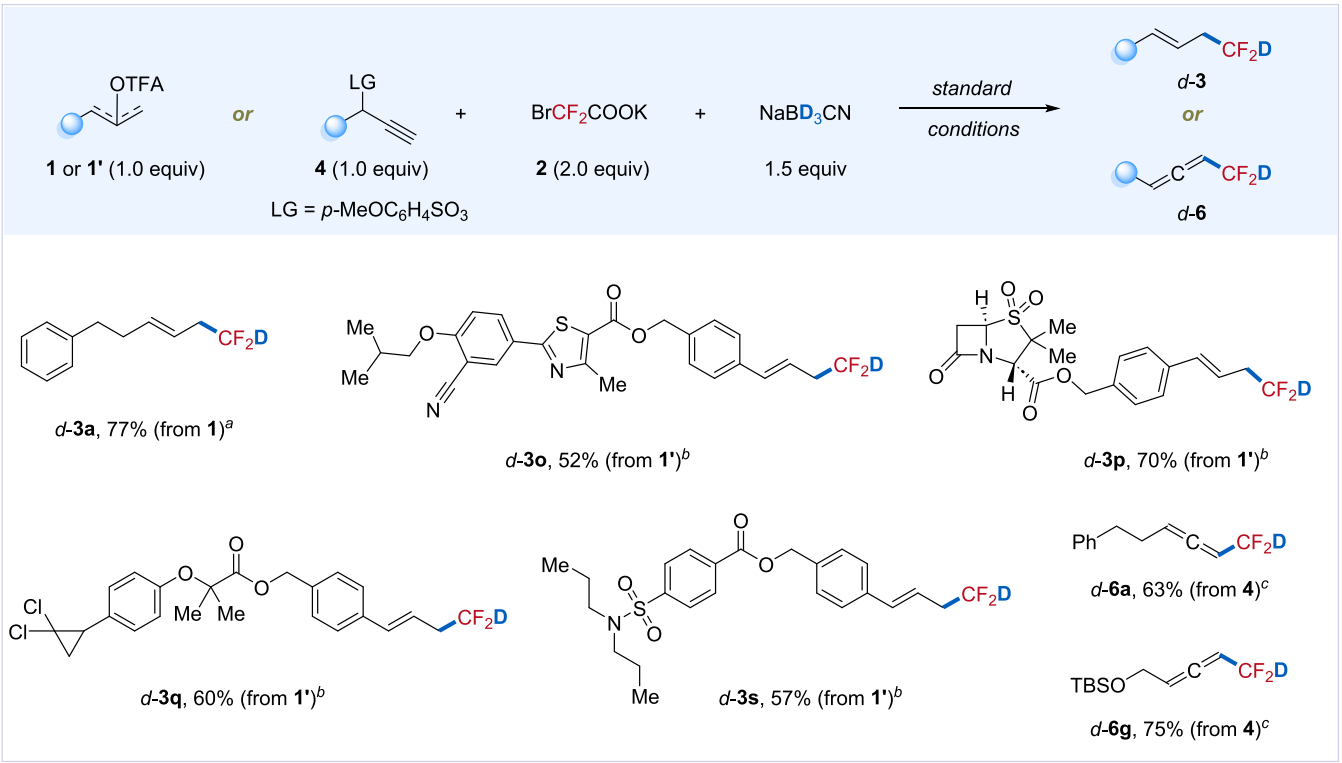
propargyl sulfonates afforded their corresponding deuterated difluoromethylated compounds efficiently. Remarkably, complex drug-like molecules were competent coupling partners, with good yields obtained (*d*-**3o**−*d*-**3q**, *d*-**3s**). To the best of our knowledge, no methods have been reported to prepare these unique structures. Since the selective introduction of deuterium into bioactive molecules has critical applications in drug discovery and development,[19] the current process offers a new opportunity to explore the biological properties of CF_2D in medicinal chemistry, thus demonstrating the advancement of this protocol further.

The synthetic utility of this approach has also been demonstrated by transformations of the resulting products (Scheme 5). A large-scale (47 mmol) synthesis of **3a** proceeded smoothly, with a comparable yield obtained, thus demonstrating the synthetic practicability and reliability of this copper-catalyzed process (Scheme 5A). Notably, even with a copper catalyst loading as low as 0.1 mol %, a gram-scale synthesis of **6a** still provided a good yield (Scheme 5B). Epoxidation of **3a** with *m*-chloroperoxybenzoic acid (mCPBA) efficiently delivered synthetically versatile synthon **9** (Scheme 5A). Treatment of **6a** with salicylaldehyde via rhodium catalysis[20a] afforded cyclized difluoromethylated product **10** that is difficult to achieve through conventional methods (Scheme 5B). The reaction of allene **7a** with benzoic acid could also be catalyzed by rhodium[20b,c] to produce isocoumarin derivative **11** efficiently (Scheme 5C).

Mechanistic Studies. Based on the reaction involving $NaBH_3CN$ as the hydride source, which was supported by the formation of deuterated difluoromethylated compounds with $NaBD_3CN$ (Scheme 4), we envisioned that three possible pathways may be involved in the generation of the key intermediate $[Cu^I]-CF_2H$ during the reaction process (Scheme 6A): (a) the $[Cu^I]-CF_2H$ species is generated by the reaction of boron hydride with $[Cu^I]=CF_2$; (b) the complexation of a copper hydride $[Cu^I]-H$ with difluorocarbene initiates the reaction, followed by migratory insertion to generate the $[Cu^I]-CF_2H$ species; (c) the direct reaction of

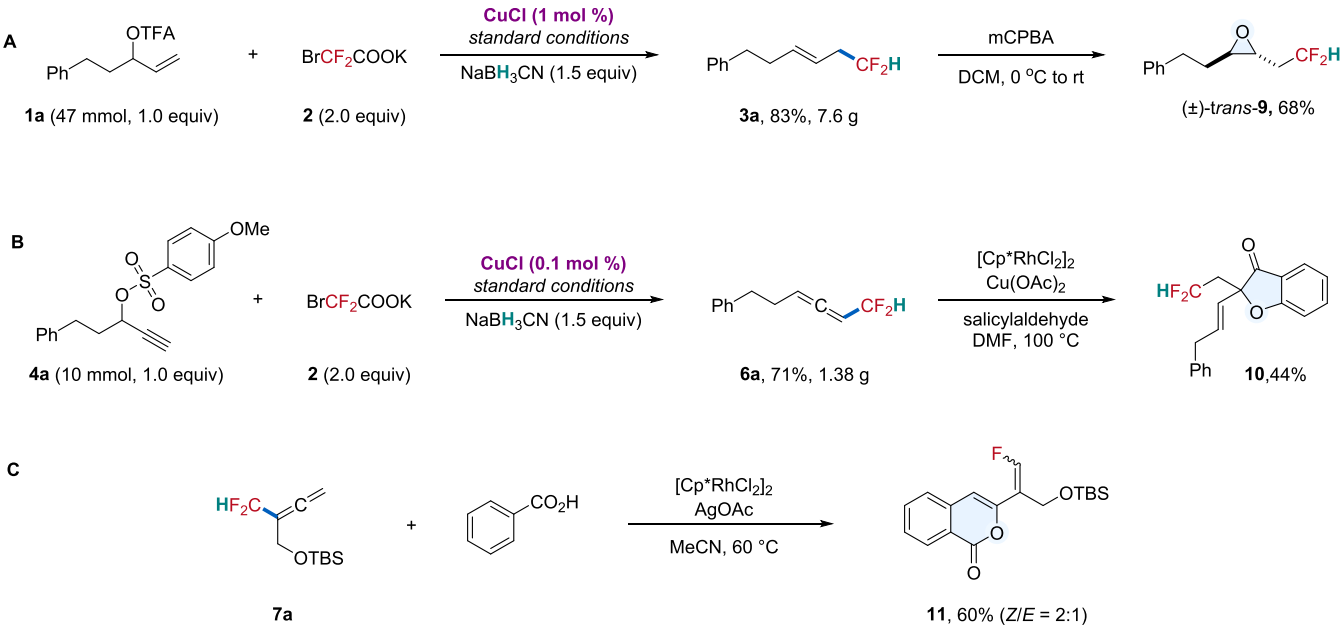
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Scheme 4. Synthesis of Deuterated Difluoromethylated Compounds[a]



[a]Condition A: **1** (0.4 mmol, 1.0 equiv), **2** (2.0 equiv), $NaBD_3CN$ (1.5 equiv), CuCl (5 mol %), **L2** (5 mol %), MeCN (8 mL). [b]Condition B: **1′**(0.4 mmol, 1.0 equiv), **2** (2.0 equiv), $NaBD_3CN$ (1.5 equiv), CuCl (5 mol %), **L7** (5 mol %), NaI (0.1 equiv), and MeCN (8 mL). [c]Condition C: **4** (0.4 mmol, 1.0 equiv), **2** (2.0 equiv), $NaBD_3CN$ (1.5 equiv), CuCl (5 mol %), **L6** (5 mol %), MeCN (4 mL).

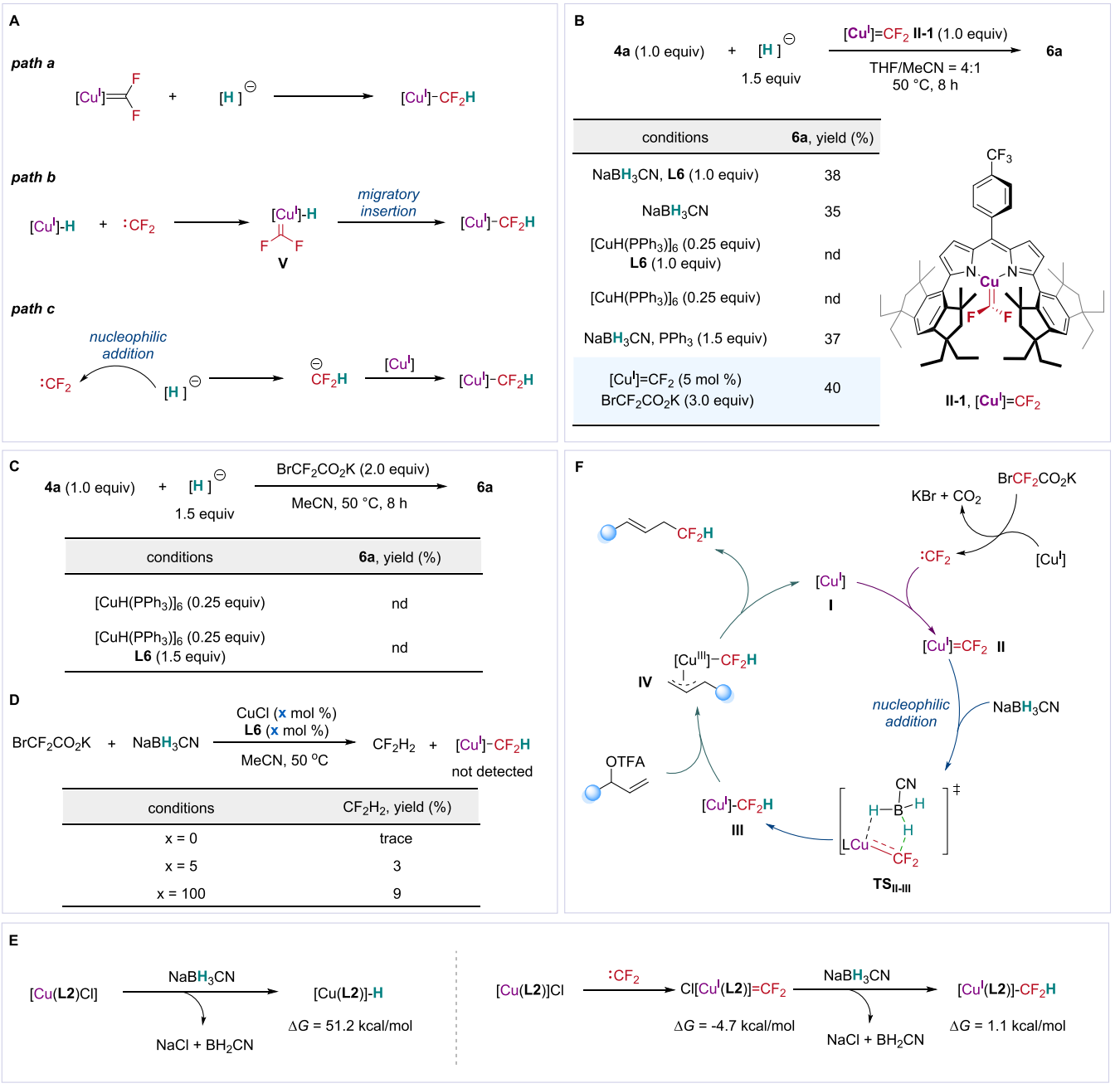
Scheme 5. Transformations of Difluoromethylated Compounds 3a, 6a, and 7a



boron hydride with difluorocarbene produces a difluoromethyl anion, which subsequently combines with [Cu^{I}] to form the [Cu^{I}]−CF_2H. Initially, several control experiments by reaction of propargyl sulfonate **4a** with $NaBH_3CN$ in the presence of [Cu^{I}]=CF_2 (**II-1**) were conducted to elucidate the possibility of path a (Scheme 6B), in which **II-1** was prepared according to our recent report.[9] We found that comparable yields (35−38%) of **6a** were obtained when the reaction was carried out with or without **L6**; even using a catalytic amount of **II-1** (5 mol %) and 3.0 equiv of $BrCF_2CO_2K$ still could provide **6a** in a similar yield (40%). However, replacing $NaBH_3CN$ with commercially available $[CuH(PPh_3)]_6$[21] failed to afford the desired product **6a**. The control experiment by adding PPh_3 to the reaction mixture with $NaBH_3CN$ as the hydride source did

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Scheme 6. Experimental Mechanistic Studies and DFT Calculations



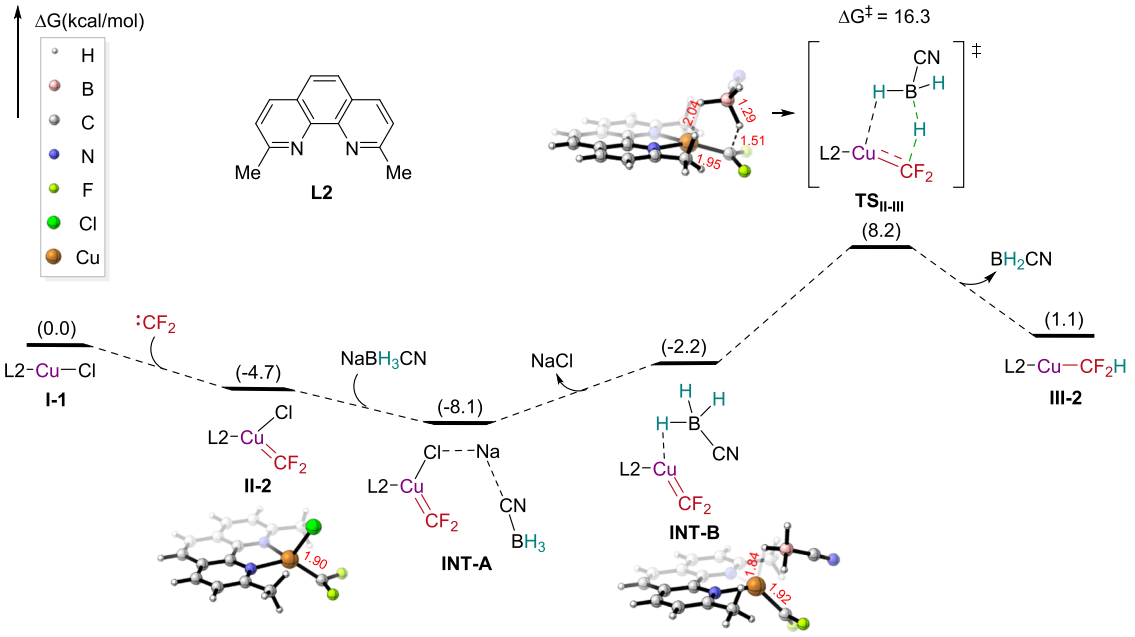
not diminish the yield, thus excluding the negative effect of PPh_3 on the reaction. These results demonstrate that the involvement of a $[Cu^I]{=}CF_2$ species in the current catalytic reaction is reasonable. The formation of $[Cu^I]$-CF_2H through path a instead of the reaction of *in situ* generated $[Cu^I]$-H with $[Cu^I]{=}CF_2$ is likely involved in this copper-catalyzed process.

Although attempts to isolate the $[Cu^I(^{EMind}L)]$-CF_2H (**III-1**) species generated between **II-1** and $NaBH_3CN$ failed due to its instability, density functional theory (DFT) calculations revealed that the formation of $[Cu^I]$-CF_2H is essentially thermoneutral (Schemes 6E and 7). As illustrated in Scheme 7, the complexation of [Cu(**L2**)Cl] (**I-1**) with difluorocarbene is 4.7 kcal/mol downhill. The resulting copper difluorocarbene complex $[ClCu^I(L2)]{=}CF_2$ (**II-2**) is then attacked by $NaBH_3CN$ via a five-membered transition state TS_{II-III} to afford difluoromethylcopper complex $[Cu^I(L2)]{-}CF_2H$ (**III-2**) with a barrier of only 16.3 kcal/mol. However, the formation of $[Cu^I(L2)]$-H between **I-1** and $NaBH_3CN$ is a thermodynamically unfavorable process (ΔG = 51.2 kcal/mol) (Scheme 6E), thus ruling out the possibility that the reaction is initiated by the formation of $[Cu^I]{-}H$, followed by the complexation with a difluorocarbene (path b). This finding was further confirmed by the reaction of $[CuH(PPh_3)]_6$ with difluorocarbene in the presence of **4a**, in which no product **6a** was observed (Scheme 6C).

We also investigated the formation of (**III-2**) by the reaction of $[Cu^I(L2)]{=}CF_2$ (**II-2**) with boron hydride to generate $[Cu^I(L2)H]{=}CF_2$ (**V-2**), followed by migratory insertion. However, attempts to optimize the structure of **V-2** failed, resulting in the formation of **III-2** (Figure S1), thus further supporting the feasibility of the formation of $[Cu^I]{-}CF_2H$ (**III**) by the nucleophilic attack of the difluorocarbene carbon

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Scheme 7. Compouted Free-Energy Profile for the Formation of [Cu^{I}(L2)]-CF$_2$H (III-2) from [Cu(L2)Cl] (I-1) Using MN15/def2-TZVP-SMD(MeCN)//MN15/def2-SVP-PCM(MeCN), Distances in Angstroms



center on [Cu^{I}(**L2**)]=CF$_2$ (**II-2**) with boron hydride. Finally, we examined the reaction of BrCF$_2$CO$_2$K with NaBH$_3$CN in the presence of CuCl/**L6** to identify the feasibility of path c (Scheme 6D). Only poor yields of CF$_2$H$_2$ and no difluoromethylcopper [Cu^{I}]-CF$_2$H were obtained when a catalytic amount of CuCl was used; even increasing the loading amount of CuCl to 1 equiv, only 9% yield of CF$_2$H$_2$ was provided without observation of [Cu^{I}]−CF$_2$H, thus excluding the involvement of path c in the reaction.

Collectively, these results demonstrate that the involvement of a [Cu^{I}]−CF$_2$H species that is generated by the reaction of NaBH$_3$CN with [Cu^{I}]=CF$_2$ is reasonable in the reaction. Accordingly, a plausible reaction mechanism was proposed (Scheme 6F). The reaction begins with the formation of the copper(I) difluorocarbene complex [Cu^{I}]=CF$_2$ (**II**) between [Cu^{I}] and difluorocarbene that is derived from the copper-assisted decomposition of BrCF$_2$CO$_2$K[9]. Subsequently, the reaction of NaBH$_3$CN with **II** produces the key intermediate [Cu^{I}]−CF$_2$H (**III**), in which the nucleophilic addition of hydride to the difluorocarbene carbon center via a five-membered transition state is involved in this process. Finally, complex **III** undergoes oxidative addition with allyl electrophiles, followed by reductive elimination to produce the final product and simultaneously release the copper(I) catalyst.

CONCLUSIONS

In summary, we have developed a new mode of catalytic difluoromethylation reaction with low-cost and readily available BrCF$_2$CO$_2$K and NaBH$_3$CN as the difluoromethyl source. The reaction allows difluoromethylation of various allyl and propargyl electrophiles with high efficiency, featuring high regio- and stereoselectivities and high functional group tolerance, even with complex drug-like molecules. The use of branched or linear propargyl electrophiles could regioselectively control the structure of difluoromethylated allenes, providing a facile route to access these valuable fluorinated structures. Remarkably, the method can also be applied to the preparation of medicinally interesting deuterated difluoromethylated compounds that otherwise are difficult to access through conventional methods. The diverse transformations of the resulting difluoromethylated products led to complex fluorinated structures, showing the prospect of the current approach in organic synthesis and medicinal chemistry. Experimental mechanistic studies and DFT calculations reveal that a [Cu^{I}]=CF$_2$ species is involved in the reaction, and the formation of the key intermediate [Cu^{I}]−CF$_2$H through the nucleophilic attack of boron hydride to [Cu^{I}]=CF$_2$ is responsible for the current copper-catalyzed process. This proof-of-concept study provides a new avenue for cost-efficiently synthesizing difluoromethylated compounds and advances the field of copper difluorocarbene chemistry.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c04949.

Experimental procedures and characterization of new compounds (PDF)

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The manuscript was written through the contributions of all authors. All authors have approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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